

OSP Cable - Optical Characteristics - G.655.C

This specification covers fibre characteristics for Coloured Non-Zero Dispersion Shifted Single-mode ITU-T Rec. G.655.C in 255 μm LT designs manufactured at the AFL Australia facility.

NZDS G655.C SINGLE-MODE FIBRE		
FIBRE SPECIFICATION (FIBRE CODE DESIGNATOR)		G.655.C (35)
STRUCTURAL		
Typical Mode Field Diameter	@1550 nm	$9.6 \pm 0.4 \mu\text{m}$
Cladding Diameter		$125 \pm 0.7 \mu\text{m}$
Max. Mode Field Concentricity Error		$0.6 \mu\text{m}$
Coating-Cladding Concentricity Error		$\leq 12 \mu\text{m}$
Cladding Non-circularity		$\leq 0.7\%$
Fibre Coating Diameter (Coloured)		$255.0 \pm 10 \mu\text{m}$
OPTICAL		
Max. Attenuation (Cabled)	@1550 nm	0.23 dB/km
	@1625 nm	0.25 dB/km
Group Refractive Index	@1550 nm	1.4691
100 Turns, 60 mm Diameter	@1550 nm	$\leq 0.05 \text{ dB}$
	@1625 nm	$\leq 0.05 \text{ dB}$
Max. Cut-off Wavelength (λ_{cc})		1450 nm
Max. Point Discontinuity at 1550 nm (Cabled)		$\leq 0.1 \text{ dB}$
Disp. Slope at 1550 nm		$\leq 0.092 \text{ (ps/nm}^2\text{.km)}$
Ch. Disp. Coefficient	@1550 – 1565 nm	$\leq 2.0 - 6.0 \text{ ps/(nm.km)}$
	@1565 – 1625 nm	$\leq 4.5 - 11.2 \text{ ps/(nm.km)}$
PMD Ind.Fibre/LDV		$\leq 0.1 / \leq 0.08 \text{ (ps/\sqrt{km})}$
MECHANICAL		
Proof Test		$\geq 1\% \text{ (100 kpsi/0.7 GPa)}$
Strip Force (F)		$1.0 \text{ N} \leq F \leq 8.9 \text{ N}$

FIBRE IDENTIFICATION	
Fibre Colour Sequence	Blue, Orange, Green, Brown, Grey, White, Red, Black, Yellow, Violet, Pink, Aqua

This datasheet is provided for guidance only and is subject to change without notice. Please ensure you have the most up to date information where parameters in this document are considered critical for your particular application.